Big Data Analytics Project 1

Social Media Engagement Analysis

Author	Syed Muhammad Waiz
Course	Big Data Analytics
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Dataset	mrudukp/social-media-sentiment-analysis (2,260 rows, 13 columns)

Key takeaway

Prioritize **TikTok** with **positive-sentiment** content, published mid-to-late week, and validate it with a controlled 4-week test. On this dataset, TikTok leads median likes (44 vs 37 for Instagram), positive posts lead negative ones by 7 median likes, and day-of-week effects are small (2 median likes).

1. Business Question

Which platforms, content sentiments, and posting days are associated with higher engagement on a ~2,000-post social-media dataset, and what should a brand test next in its publishing strategy?

This project demonstrates the full Big Data Analytics workflow: **load, inspect, clean, EDA, interpret, scale, justify**. It does not build a production model.

2. Dataset and Provenance

- **Source:** mrudukp/social-media-sentiment-analysis
- **Permission:** granted by the repository author on 2026-10-25 (screenshot included in data/permission_proof.png)
- **Shape:** 2,260 rows x 13 columns
- **Columns:** post_id, timestamp, platform, user, followers_count, verified, text, hashtags, country, likes, retweets_shares, comments, sentiment

Provenance note: The dataset is multi-platform, multi-country social-media posts. It is **not** verified as Pakistani clothing-brand data. All findings describe this dataset only and are not claims about Pakistan.

3. Data Quality and Cleaning (NB01)

Five quality issues were identified and documented:

1. **Duplicate post_ids:** 44 fully identical rows and 60 duplicated post_id values (16 partial duplicates: rows sharing an ID but with different content). *Decision:* keep partial duplicates (they may be legitimate cross-platform re-posts); drop only fully identical rows.
2. **Platform case inconsistency:** "TWITTER" vs "Twitter", etc. 10 raw values collapse to 5 clean platforms. Without cleaning, roughly 10-15% of rows would be misclassified.
3. **Country case and whitespace inconsistency:** "usa " vs "USA", etc. 24 raw values collapse to 12 clean countries.
4. **Numeric columns stored as strings** (comma-formatted like "1,125"), converted to float64 via pd.to_numeric(errors="coerce").
5. **Impossible negative like value** (likes = -1), treated as a sentinel value and converted to NaN.

Missing values

Missing values were documented, not auto-dropped:

Column	Nulls	Interpretation
hashtags	788 (34.87%)	**Legitimate blank**: post had no hashtag
country	126 (5.58%)	True missing
followers_count	90 (3.98%)	True missing
comments	46 (2.04%)	True missing (not zero)
verified	44 (1.95%)	True missing
sentiment	33 (1.46%)	True missing

All four engagement metrics (likes, shares, comments, followers) are **right-skewed** (skewness > 1). Therefore the **median** is used throughout NB02-NB04 for group comparisons; the mean would be inflated by a handful of high-engagement posts.

4. EDA and Findings (NB02)

After cleaning and re-deduplication, the final dataset contains **2,200 rows**.

Finding 1: TikTok leads median engagement

Platform	n	Median likes	Median shares	Median comments
TikTok	390	**44.0**	**8.0**	**4.0**
Instagram	595	37.0	6.0	3.0
Twitter	726	32.0	5.0	3.0
Facebook	370	26.0	5.0	2.0
LinkedIn	119	20.0	3.0	2.0

TikTok posts receive a median of **+7 likes, +3 shares, +2 comments** over Instagram, the next-best platform by median likes.

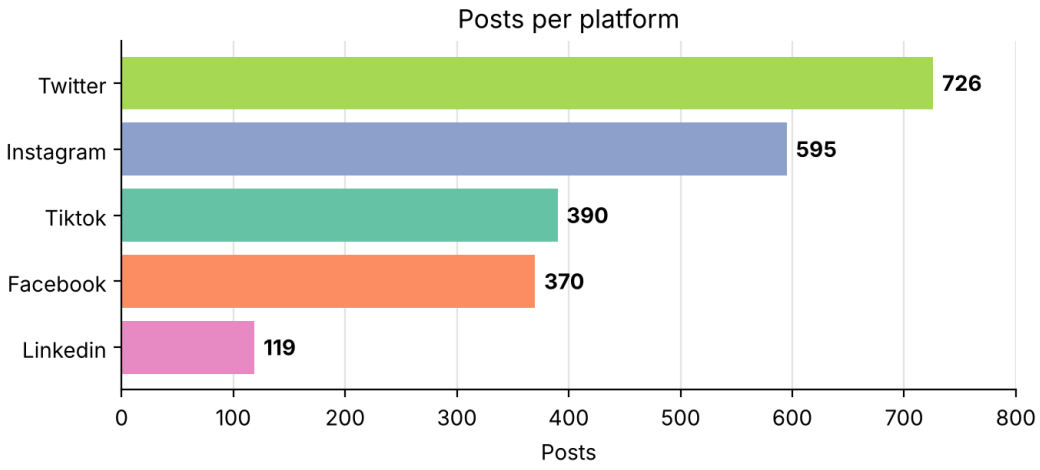


Figure 1. Posts per platform (volume)

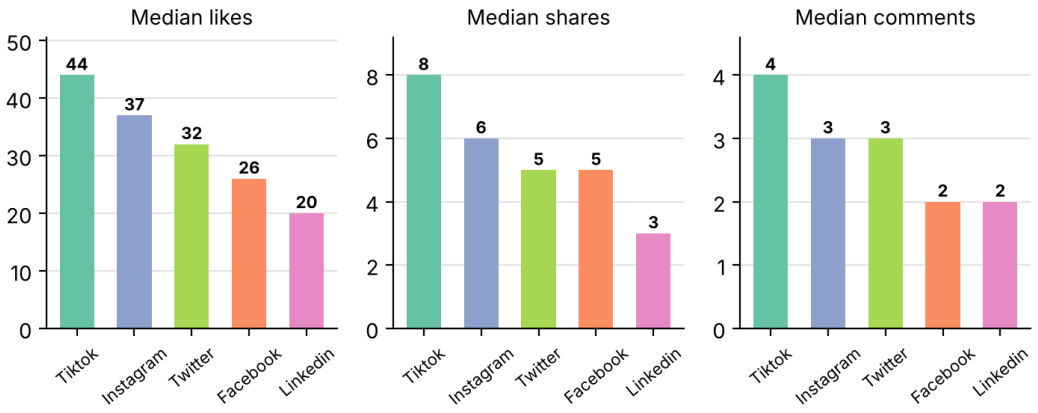


Figure 2. Median engagement by platform (likes / shares / comments)

Finding 2: Positive sentiment gets more engagement

Sentiment	n	Median likes	Median shares
Positive	879	**39.0**	**7.0**
Neutral	652	28.0	4.5
Negative	636	32.0	5.0

Positive-sentiment posts receive **+7 median likes** and **+2 median shares** over negative-sentiment posts. Neutral posts are the lowest, suggesting that *clear* emotional framing (either direction) outperforms neutral messaging, with positive winning.

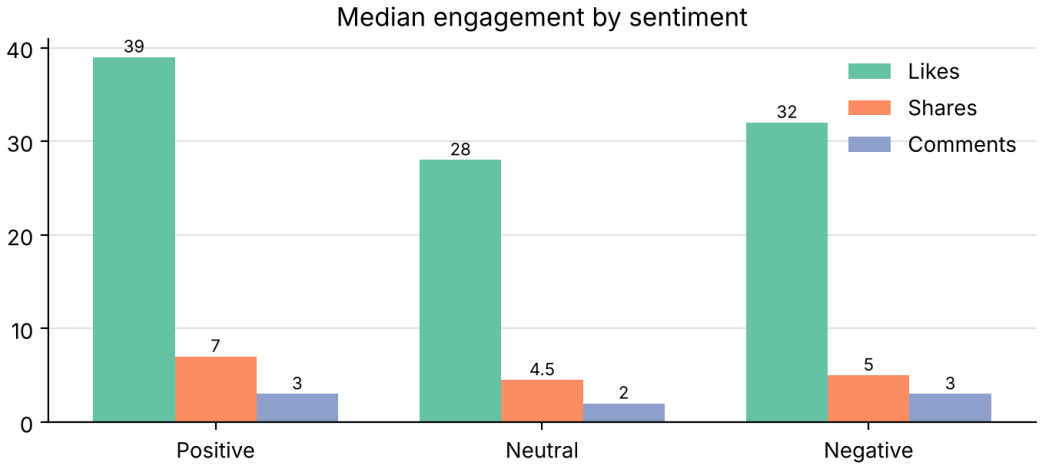


Figure 3. Median engagement by sentiment

Finding 3: Thursday, Friday and Sunday lead

Day	n	Median likes
Thursday	333	**34.0**
Friday	339	34.0
Sunday	267	34.0
Wednesday	291	33.0
Saturday	306	33.0
Monday	326	32.0
Tuesday	338	32.0

The gap between the best and worst day is only **2 median likes**: small, but Thursday/Friday/Sunday consistently outperform Monday/Tuesday. The pattern is directional, not dramatic.

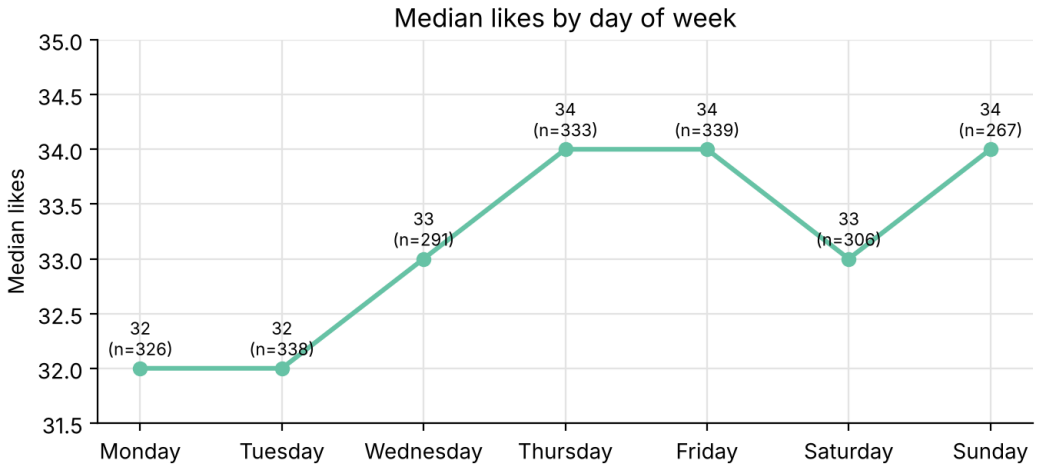


Figure 4. Median likes by day of week

Finding 4: Follower count is a moderate predictor of likes

The log-log Pearson correlation between followers and likes is **r = 0.493** (moderate positive). Audience size explains roughly **r^2 = 24%** of the variance in likes, so content and platform choice account for much of the remainder. **Audience size alone does not determine engagement.**

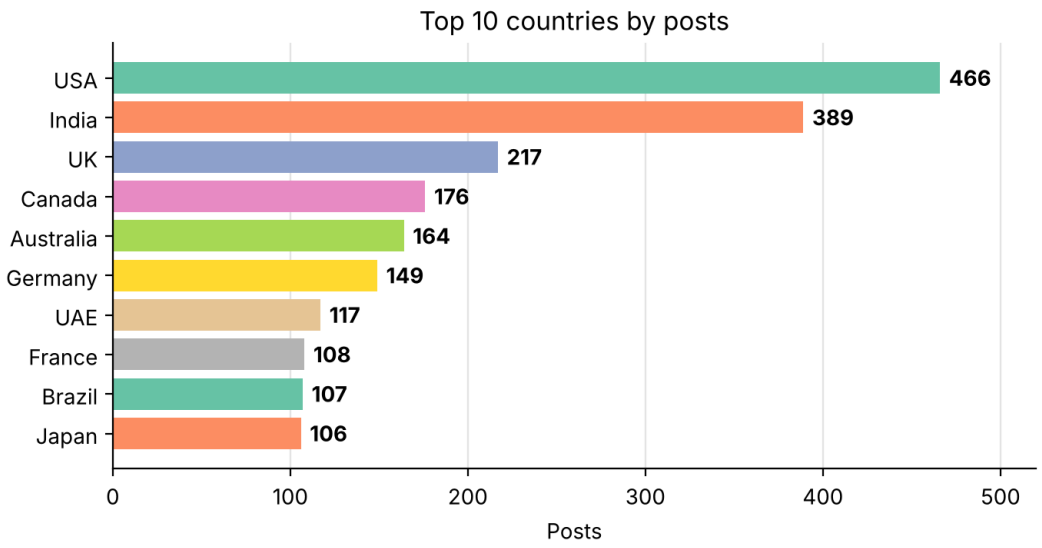


Figure 5. Top 10 countries by posts

5. Business Recommendation (NB03)

> **Prioritize TikTok with positive-sentiment content, published mid-to-late week, and measure the result with a controlled 4-week test.**

Supporting evidence

1. **Platform:** TikTok median **44 likes** per post (n=390) vs Instagram **37** (n=595). Delta = +7 likes. The sample is large enough to trust the direction.
2. **Sentiment:** Positive median **39 likes** vs Negative **32** and Neutral **28**. Delta (Positive - Negative) = +7 likes. Neutral is the worst performer, so *emotional clarity* matters more than positivity alone.
3. **Timing:** Thursday / Friday / Sunday show median **34 likes** vs Monday / Tuesday at **32**. The effect is small (2 likes) but directionally consistent.

Action plan

- **Weeks 1-4:** Publish 1 TikTok-native post per week with positive tone, on Thursday. Publish a matched control post on Instagram the same week.
- **Measurement:** Compare median likes across the 4-week window. Track reach if available (currently missing from the dataset).
- **Guardrail:** Do not compare total likes. Use **median** and **per-post** metrics, because the data is right-skewed.

Limitations of this recommendation

- No reach or impressions, so **engagement rate** cannot be computed; only raw counts are available.
- followers_count is missing for **88 posts**: audience-controlled comparison is possible for about 96% of the data but not all.
- **Sentiment labels are pre-supplied** by the dataset and their provenance is unknown, so "positive sentiment leads to more likes" should be re-verified on the brand's own labelled content.
- The dataset is **not Pakistani clothing-brand data**: these findings are directional and should be re-validated on the brand's own account.

6. Scaling Experiment and Tool Justification (NB04)

The same groupby("platform").median() pipeline was benchmarked at increasing dataset sizes by concatenating the cleaned 2,200-row dataset N times. **This is a scaling test only: the repeated rows are not new real-world evidence.**

Results (measured on Google Colab)

Target rows	Actual rows	Build (s)	Groupby (s)	Memory (MB)
2,000	2,200	0.000	**0.024**	1.4
20,000	19,800	0.002	**0.005**	12.9
200,000	198,000	0.034	**0.036**	128.7
2,000,000	1,999,800	0.308	**0.355**	1,299.7

Groupby wall-clock time scales roughly linearly up to 2 million rows, and memory grows linearly at approximately **650 bytes per row** (1.3 GB at 2M rows). Even at 2M rows, a single Pandas groupby runs in **0.355 seconds**, well within interactive tolerance.

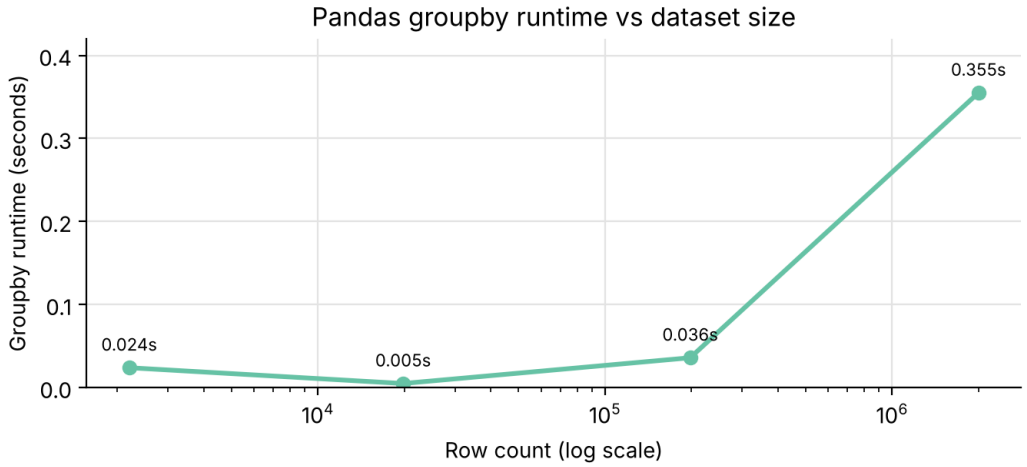


Figure 6. Scaling benchmark: groupby runtime vs row count

Would Spark help? Measured, not guessed

- **At 2M rows (measured): NO.** Pandas completes in 0.355 s on a single Colab VM. Spark's JVM startup (~10-20 s), shuffle overhead, and cluster-management cost would exceed the compute saved.
- **At 50M+ rows (extrapolated from the linear trend): LIKELY YES.** Reasons:

a **Distributed storage** (HDFS/S3) avoids needing the whole dataset in RAM (2M rows = 1.3 GB; 50M rows is about 32 GB, exceeding typical single-node RAM).

b **Parallel groupby** across machines cuts wall-clock time.

c **Fault tolerance** during multi-hour jobs.

d **Cross-team access** through shared storage and a SQL-like interface.

Conclusion: For this project's 2,200-row dataset, Pandas is strictly correct. Spark becomes a candidate tool when data volume, latency, or distributed-storage requirements actually justify the added operational complexity, not by default for any dataset labeled "big."

7. Analytics Preview (Conceptual)

The dataset supports three natural follow-up modeling paths. Each is **previewed conceptually**, not built in Project 1.

Technique	Approach	Notes
Classification	Predict "high vs low engagement" using a median-split label (not the target metric itself).	Features: platform, sentiment, has_hashtag, followers_count. Train/test split mandatory. Output: confusion matrix + accuracy.
Clustering	Group posts by standardized (likes, shares, comments) to discover engagement archetypes.	Start with k=3; interpret centers as low / mid / high engagement segments.
Regression	Predict likes from platform, sentiment, and log(followers). Use log(likes) as target given right-skew.	Train/test split mandatory. **Strongest fit**: quantifies which factors predict engagement.

8. Limitations (Consolidated)

1. No reach or impressions, so **engagement rate is not computable**.
2. followers_count missing for **88 posts** (4% of cleaned data).
3. **Sentiment labels** are pre-supplied by the dataset; provenance unknown.
4. The dataset is **not Pakistan-specific**: findings are directional and should be re-validated on the brand's own data.
5. Only 2,200 cleaned rows: adequate for EDA but not for training a reliable predictive model. Large-scale ML would need 50k+ rows.
6. Scaling results are measured on Colab hardware (single VM) and should be re-benchmarked on any deployment target.

9. Conclusion

A 2,260-row social-media dataset was cleaned (case normalization, deduplication, sentinel-value handling), explored with median-based comparisons across platform / sentiment / country / day-of-week, and benchmarked at up to 2 million rows to establish where Pandas suffices and where Spark would help. The data suggests prioritizing **TikTok**, **positive sentiment**, and **Thursday** as the highest-leverage combination. A 4-week controlled test would convert these dataset-level patterns into brand-level validation, and adding paid-campaign data (spend, impressions, clicks) would enable ROAS analysis in Project 2.

Exit question: Which technique best answers the marketing question?
Regression, because "how many likes can we expect from this post, given platform + sentiment + audience?" is a numerical prediction problem that directly informs content-investment decisions.